

# Statistical Inference

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## **Abstract**

Notes for the basics of Probability Theory.

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# 1 Fundamentals of Statistics

## 1.1 Random Samples and Statistics

### Definition 1.1.1 (Random Samples)

Let  $(\Omega, \mathcal{F}, P)$  be a probability space. Let  $X_1, \dots, X_n : \Omega \rightarrow \mathbb{R}$  be random variables. We say that they are a random sample of size  $n$  from the population  $f$  if the following are true.

- $X_1, \dots, X_n$  are pairwise independent.
- The probability density (mass) function is given by  $f_{X_i} = f_X$ .

The fact that their probability density functions are equal imply that the random variables are identically distributed.

### Definition 1.1.2 (Statistics)

Let  $(\Omega, \mathcal{F}, P)$  be a probability space. Let  $X_1, \dots, X_n : \Omega \rightarrow \mathbb{R}$  be a random sample. A statistic of the random sample is a random variable  $T(X_1, \dots, X_n)$  for some function  $T : \mathbb{R}^n \rightarrow \mathbb{R}$ .

### Definition 1.1.3 (Sample Mean)

Let  $(\Omega, \mathcal{F}, P)$  be a probability space. Let  $X_1, \dots, X_n : \Omega \rightarrow \mathbb{R}$  be a random sample. Define the sample mean of the random sample to be the statistic defined by

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

### Definition 1.1.4 (Sample Variance)

Let  $(\Omega, \mathcal{F}, P)$  be a probability space. Let  $X_1, \dots, X_n : \Omega \rightarrow \mathbb{R}$  be a random sample. Define the sample variance of the random sample to be the statistic defined by

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

### Definition 1.1.5 (Sample Standard Deviation)

Let  $(\Omega, \mathcal{F}, P)$  be a probability space. Let  $X_1, \dots, X_n : \Omega \rightarrow \mathbb{R}$  be a random sample. Define the sample standard deviation of the random sample to be the statistic defined by

$$S = \sqrt{S^2}$$

## 2 Principles of Data Reduction

### 2.1 Sufficiency Principle

Suppose that  $(\Omega, \mathcal{F}, P)$  is a probability space. Often we do not know what  $P$  is actually like. To probe the properties of  $P$ , we may use statistics  $T(X_1, \dots, X_n)$  on the probability space.

A sufficient statistic  $T$  is a statistic where, informally speaking, all information about  $P$  must come from  $T$ .

#### Definition 2.1.1 (Sufficient Statistics)

Let  $(\Omega, \mathcal{F}, P)$  be a collection of probability spaces for  $\theta \in I$ . Let  $\{D_\theta \mid \theta \in I\}$  be a class of distributions. Let  $X_\theta : \Omega \rightarrow \mathbb{R}^n$  be a random sample with distribution  $D_\theta$ . Let  $T$  be a statistic of the random sample. We say that  $T$  is sufficient if the conditional probability density function  $f_{X_\theta|T}$  is a constant function with respect to  $\theta$ .

#### Example 2.1.2

Let  $(\Omega, \mathcal{F}, P)$  be a probability space. Let  $X_1, \dots, X_n : \Omega \rightarrow \mathbb{R}$  be a random sample of Bernoulli distribution with parameter  $\theta \in (0, 1)$  (Note that for each  $\theta$ ,  $P_{X_i} = f_{X_i}$  is different since the formula for  $f_{X_i}$  depends on  $\theta$ ). Then  $T = \sum_{i=1}^n X_i$  is a sufficient statistic for  $\theta$ .

#### Proof

Let  $\theta \in \mathbb{R}$ . We use the above lemma. We have

$$\begin{aligned} P_\theta(\mathbf{X} = \mathbf{x} \mid T(\mathbf{X}) = T(\mathbf{x})) &= \frac{f_{X_1, \dots, X_n}(x_1, \dots, x_n)}{f_{T(X_1, \dots, X_n)}(x_1, \dots, x_n)} \\ &= \frac{\prod_{i=1}^n f_{X_i}(x_i)}{f_{X_1 + \dots + X_n}(x_1, \dots, x_n)} \\ &= \frac{\prod_{i=1}^n \theta^{x_i} (1 - \theta)^{1-x_i}}{\binom{n}{\sum_{i=1}^n x_i} \theta^{\sum_{i=1}^n x_i} (1 - \theta)^{n - \sum_{i=1}^n x_i}} \\ &= \frac{\theta^{\sum_{i=1}^n x_i} (1 - \theta)^{\sum_{i=1}^n x_i}}{\binom{n}{\sum_{i=1}^n x_i} \theta^{\sum_{i=1}^n x_i} (1 - \theta)^{n - \sum_{i=1}^n x_i}} \\ &= \frac{1}{\binom{n}{\sum_{i=1}^n x_i}} \end{aligned}$$

■

#### Theorem 2.1.3 (Neyman-Fisher Factorization Theorem)

Let  $(\Omega, \mathcal{F}, P)$  be a probability space. Let  $\{D_\theta \mid \theta \in I\}$  be a family of distributions. Let  $X_\theta : \Omega \rightarrow \mathbb{R}^n$  be a random sample with distribution  $D_\theta$ . Let  $T$  be a statistic of the random sample. Let  $f_\theta$  be the probability density function of  $X_\theta$ . Then  $T$  is sufficient if and only if there exists probability density functions  $g_\theta$  for each  $\theta$  such that

$$f_\theta(x) = g_\theta(T(x))h(x)$$

for all  $x \in \Omega$  and all  $\theta \in I$ .

#### Definition 2.1.4 (Minimal Sufficient Statistics)

Let  $(\Omega, \mathcal{F}, P)$  be a probability space. Let  $\{D_\theta \mid \theta \in I\}$  be a family of distributions. Let  $X_\theta : \Omega \rightarrow \mathbb{R}^n$  be a random sample with distribution  $D_\theta$ . Let  $T$  be a sufficient statistic of the random sample. We say that  $T$  is minimal if for any other sufficient statistic  $T'$ ,  $T'(x) = T'(y)$  implies  $T(x) = T(y)$ .

#### Definition 2.1.5 (Ancillary Statistics)

Let  $(\Omega, \mathcal{F}, P)$  be a probability space. Let  $\{D_\theta \mid \theta \in I\}$  be a family of distributions. Let  $X_\theta : \Omega \rightarrow \mathbb{R}^n$

be a random sample with distribution  $D_\theta$ . Let  $T$  be a statistic of the random sample. We say that  $T$  is ancillary if  $T$  is a constant function with respect to  $\theta$ .

### Definition 2.1.6 (Complete Statistics)

### Theorem 2.1.7 (Basu's Theorem)

Let  $(\Omega, \mathcal{F}, P)$  be a probability space. Let  $\{D_\theta \mid \theta \in I\}$  be a family of distributions. Let  $X_\theta : \Omega \rightarrow \mathbb{R}^n$  be a random sample with distribution  $D_\theta$ . Let  $T$  be a statistic of the random sample. If  $T$  is complete and minimal sufficient, then  $T$  is independent of any ancillary statistic.

## 2.2 Likelihood Principle

### Definition 2.2.1 (The Likelihood Function)

Let  $(\Omega, \mathcal{F}, P)$  be a probability space. Let  $\{D_\theta \mid \theta \in I\}$  be a family of distributions. Let  $X_\theta : \Omega \rightarrow \mathbb{R}^n$  be a random sample with distribution  $D_\theta$ . Define the likelihood function  $L : I \rightarrow \{g : \mathbb{R} \rightarrow \mathbb{R} \mid g \text{ is a function}\}$  of the random sample to be

$$L(\theta) = f_\theta$$

where  $f_\theta$  is the probability density (mass) function of  $X_\theta$ .

## 2.3 Conditionality Principle

## 2.4 Equivariance Principle

### 3 Estimators

Suppose some real life problem is modelling statistically by the probability space  $(\Omega, \mathcal{F}, P)$ . To probe what  $P$  looks like, we sample the space using random variables  $X_1, \dots, X_n : \Omega \rightarrow \mathbb{R}$ . Based on the sample, we may infer that  $P$  may have parameter  $\hat{\theta}$  (naturally  $\hat{\theta}$  should be a function of  $X_1, \dots, X_n$ ). We may then need to evaluate how accurate  $\hat{\theta}$  is compared to the actual parameter  $\theta$ .

There are two main branches in estimator theory: We need different methods for finding estimators, and we need to evaluate how accurate the estimators are.

There are also two types of estimators: Point estimators and interval estimators.

#### 3.1 Finding Point Estimators

##### Definition 3.1.1 (Point Estimators)

Let  $(\Omega, \mathcal{F}, P)$  be a probability space. Let  $I \subseteq \mathbb{R}$  be an indexing set. Let  $\{D_\theta \mid \theta \in I\}$  be a family of distributions. Let  $X_\theta : \Omega \rightarrow \mathbb{R}^n$  be a random sample with distribution  $D_\theta$ . A point estimator of the random sample is a function  $\hat{\theta} \circ X_\theta : \Omega \rightarrow I$ .

Note: This the same definition of a statistic.

##### Definition 3.1.2 (Method of Moments)

Let  $(\Omega, \mathcal{F}, P)$  be a probability space. Let  $J_1, \dots, J_k \subseteq \mathbb{R}$  be indexing sets. Let  $\{D_{\theta_1, \dots, \theta_k} \mid \theta_i \in J_i\}$  be a family of distributions. Let  $X_{\theta_1, \dots, \theta_k} : \Omega \rightarrow \mathbb{R}^n$  be a random sample with distribution  $D_{\theta_1, \dots, \theta_k}$  and probability distribution function  $f_{\theta_1, \dots, \theta_k}$ . Define the methods of moments estimator to be the statistics  $\hat{\theta}_1, \dots, \hat{\theta}_k$  given by the solving the system of equations

$$\frac{1}{n} \sum_{j=1}^n (X_{\theta_1, \dots, \theta_k})_j^i = E[D_{\theta_1, \dots, \theta_k}^i]$$

for  $1 \leq i \leq k$ .

##### Example 3.1.3

Let  $(\Omega, \mathcal{F}, P)$  be a probability space. Let  $X : \Omega \rightarrow \mathbb{R}^n$  be a random sample with normal distribution  $X_i \sim \text{Normal}(\theta_1, \theta_2)$ . The method of moments estimator for  $X$  is given by solving

$$\frac{1}{n} \sum_{j=1}^n (X_{\theta_1, \dots, \theta_k})_j = E[\text{Normal}(\hat{\theta}_1, \hat{\theta}_2)] \quad \text{and} \quad \frac{1}{n} \sum_{j=1}^n (X_{\theta_1, \dots, \theta_k})_j^2 = E[\text{Normal}(\hat{\theta}_1, \hat{\theta}_2)^2]$$

which we may rewrite as

$$\overline{X_{\theta_1, \dots, \theta_k}} = \hat{\theta}_1 \quad \text{and} \quad \frac{1}{n} \sum_{j=1}^n (X_{\theta_1, \dots, \theta_k})_j^2 = \hat{\theta}_1^2 + \hat{\theta}_2^2$$

Omitting the dependency of the random variables on the parameters, we get

$$\hat{\theta}_1 = \bar{X} \quad \text{and} \quad \hat{\theta}_2 = \frac{1}{n} \sum_{j=1}^n (X_j - \bar{X})^2$$

We are lucky in the above example because the estimator accurately recovers the actual parameters. But in general, this is not true.

**Example 3.1.4**

Let  $(\Omega, \mathcal{F}, P)$  be a probability space. Let  $X : \Omega \rightarrow \mathbb{R}^n$  be a random sample with binomial distribution  $X_i \sim \text{Binom}(\theta_1, \theta_2)$ . The method of moments estimator for  $X$  is given by solving

$$\frac{1}{n} \sum_{j=1}^n X_j = E[\text{Binom}(\hat{\theta}_1, \hat{\theta}_2)] \quad \text{and} \quad \frac{1}{n} \sum_{j=1}^n X_j^2 = E[\text{Binom}(\hat{\theta}_1, \hat{\theta}_2)^2]$$

which we may rewrite as

$$\bar{X} = \hat{\theta}_1 \hat{\theta}_2 \quad \text{and} \quad \frac{1}{n} \sum_{j=1}^n X_j^2 = \hat{\theta}_1 \hat{\theta}_2 (1 - \hat{\theta}_2) + \hat{\theta}_1^2 \hat{\theta}_2^2$$

Solving the simultaneous equation gives

$$\hat{\theta}_1 = \frac{\bar{X}^2}{\bar{X} - \frac{1}{n} \sum_{j=1}^n (X_j - \bar{X})^2} \quad \text{and} \quad \hat{\theta}_2 = \frac{\bar{X} - \frac{1}{n} \sum_{j=1}^n (X_j - \bar{X})^2}{\bar{X}} = \frac{\bar{X}}{\hat{\theta}_1}$$

**Definition 3.1.5 (Maximum Likelihood Estimators)**

Let  $(\Omega, \mathcal{F}, P)$  be a probability space. Let  $I \subseteq \mathbb{R}$  be an indexing set. Let  $\{D_\theta \mid \theta \in I\}$  be a family of distributions. Let  $X_\theta : \Omega \rightarrow \mathbb{R}^n$  be a random sample with distribution  $D_\theta$ . Let  $\hat{\theta}$  be a point estimator for the random sample. We say that  $\hat{\theta}$  is a maximum likelihood estimator if there exists  $x \in \mathbb{R}^n$  such that

$$L(\hat{\theta}(x))(x) = \max\{L(\theta)(x) : \theta \in I\}$$

**3.2 Evaluating Point Estimators****Definition 3.2.1 (Bias of a Point Estimator)**

Let  $(\Omega, \mathcal{F}, P)$  be a probability space. Let  $X_1, \dots, X_n : \Omega \rightarrow \mathbb{R}$  be a random sample. Let  $\hat{\theta}$  be a point estimator of the random sample. Define the bias of  $\hat{\theta}$  to be

$$\text{Bias}(\hat{\theta}) = E[\hat{\theta}] - \theta$$

**Definition 3.2.2 (Unbiased Point Estimators)**

Let  $(\Omega, \mathcal{F}, P)$  be a probability space. Let  $X_1, \dots, X_n : \Omega \rightarrow \mathbb{R}$  be a random sample. Let  $\hat{\theta}$  be a point estimator of the random sample. We say that  $\hat{\theta}$  is unbiased if  $\text{Bias}(\hat{\theta}) = 0$ .

**Definition 3.2.3 (Mean Squared Error)**

Let  $(\Omega, \mathcal{F}, P)$  be a probability space. Let  $X_1, \dots, X_n : \Omega \rightarrow \mathbb{R}$  be a random sample. Let  $\hat{\theta}$  be a point estimator of the random sample. Define the mean squared error of  $\hat{\theta}$  to be

$$\text{MSE}(\hat{\theta}) = E[(\hat{\theta} - \theta)^2]$$

**Theorem 3.2.4 (The Bias Variance Trade Off)**

Let  $(\Omega, \mathcal{F}, P)$  be a probability space. Let  $X_1, \dots, X_n : \Omega \rightarrow \mathbb{R}$  be a random sample. Let  $\hat{\theta}$  be a point estimator of the random sample. Then we have

$$\text{MSE}(\hat{\theta}) = \text{Var}(\hat{\theta}) + (\text{Bias}(\hat{\theta}))^2$$

**3.3 Finding Interval Estimators****3.4 Evaluating Interval Estimators**

## 4 Hypothesis Testing

### 4.1 Finding Hypothesis Tests

**Definition 4.1.1** (Hypothesis)

Let  $(\Omega, \mathcal{F}, P)$  be a probability space. Let  $\{D_\theta \mid \theta \in I\}$  be a family of distributions. Let  $X_\theta : \Omega \rightarrow \mathbb{R}^n$  be a random sample with distribution  $D_\theta$ . A hypothesis is a subset of  $I$ .

### 4.2 Evaluating Hypothesis Tests



## 5 Regression Analysis

### Definition 5.0.1 (Linear Regression Space)

Let  $(\Omega, \mathcal{F}, P)$  be a probability space. Let  $X : \Omega \rightarrow \mathbb{R}^n$  be a random sample such that  $E[X] < \infty$ . Define the linear regression space of the sample to be

$$\mathcal{H}(X) = \left\{ \sum_{i=1}^n \beta_i X_i + \alpha \mid \alpha, \beta_i \in \mathbb{R} \right\} \subseteq L^2(\Omega)$$

### Definition 5.0.2 (Linear Regression)

Let  $(\Omega, \mathcal{F}, P)$  be a probability space. Let  $X : \Omega \rightarrow \mathbb{R}^n$  be a random sample. Let  $Y : \Omega \rightarrow \mathbb{R}^n$  be a random sample. Let  $L : \mathcal{H} \rightarrow \mathbb{R}$  be a loss function. The linear regression of  $Y$  on  $X$  with  $L$  is a random variable  $\hat{Y} \in \mathcal{H}(X)$  such that

$$L(\hat{Y}) = \min\{L(Z) \mid Z \in \mathcal{H}(X)\}$$

### Proposition 5.0.3

Let  $(\Omega, \mathcal{F}, P)$  be a probability space. Let  $X : \Omega \rightarrow \mathbb{R}^n$  be a random sample. Let  $Y : \Omega \rightarrow \mathbb{R}^n$  be a random sample. Then the linear regression of  $Y$  on  $X$  with the mean squared error exists and is unique.